Sum and difference of cubes

1 If $a^3 + 125 = (a+5)\left(a^2 + ka + 25\right)$, find the value of k.

2 Factor:

a $x^3 + y^3$ b $x^3 - y^3$ c $x^3 + 1$ d $x^3 - 8$

e $x^3 + 27$ f $x^3 - 64$ g $x^3 + 512$ h $x^3 - 1000$

3 Factor:

a $x^3 + 64y^3$ b $x^3 - 125y^3$ c $x^3 + 729y^3$ d $x^3 - 216y^3$

e $x^3 + 1000y^3$ f $a^3 + 27b^3$ g $s^3 + 125t^3$ h $u^3 - 343v^3$

4 Factor:

a $27x^3 + 64$ b $27x^3 - 8$

c $1000x^3 + 729$ d $512x^3 - 343$

e $8x^3 + 1000$ f $125x^3 - 1$

g $512x^3 + 8$ h $216x^3 - 64$

5 Factor:

a $250u^3 + 54v^3$ b $54m^3 - 250n^3$

c $27x^3 + 8y^3$ d $125x^3 - 64y^3$

e $216x^3 + 343y^3$ f $343x^3 - 1000y^3$

g $40u^3 + 5v^3$ h $4m^3 - 32n^3$

6 Factor:

a $(x+7)^3 + (x-8)^3$ b $(x-3)^3 - (x+2)^3$

c $(x-2)^3 + (x+5)^3$ d $(x+1)^3 - (x-7)^3$

7 Simplify the following:

a $\dfrac{125x^3 + 8}{5x + 2}$ b $\dfrac{x^3 - 125}{4x - 20}$

c $\dfrac{x^3 - 125}{x^2 - 25}$ d $\dfrac{8x^3 - 27y^3}{2x - 3y}$

8 Consider the polynomial $x^6 - y^6$.

a By thinking of the polynomial as $\left(x^3\right)^2 - \left(y^3\right)^2$, fully factor it starting with a difference of two squares.

b By thinking of the polynomial as $\left(x^2\right)^3 - \left(y^2\right)^3$, show that we can use a difference of two cubes to write it in the form $(x-y)(x+y)\left(x^4 + x^2y^2 + y^4\right)$.

c By comparing the results of parts (a) and (b), write down a factoring for $x^4 + x^2y^2 + y^4$.

9 Evaluate $11^3 + 6^3$ by factorising it using the sum of two cubes method.